0582

National University of Computer and Emerging Sciences, Lahore Campus

Course:	COAL	Course Code:	EE2003
Program:	BS(CS,SE)	Semester:	Fall 2024
Duration:	1 Hour	Total Marks:	40
Paper Date:	24-09-2024	Page(s):	4
Section:	All	Roll No.	Solution
Exam:	Midterm-I	Your Section:	

Instruction/Notes: This is an open book exam. Sharing book or calculators is NOT ALLOWED. PAPERS/NOTES ATTACHED WITH BOOK ARE NOT ALLOWED. Rough sheets can be used but will not be collected and checked. In case of any ambiguity, make reasonable assumptions. Questions during exams are not allowed.

Question 1 [CLO 1] [10 Marks]: Short questions

i) [1 Mark] To address 1-MB of memory we need 20 bits of addressing, 1024 MB can be accessed using 30 bits of addressing.

ii) [2 Mark] For each of the instructions given below, **identify whether the instruction is valid or invalid:**

Instruction	Valid/Invalid	Instruction	Valid/Invalid
mov [bx+si+di],10	Invalid	mov word [mem1],al	Invalid
add word[mem1], word[mem2]	Invalid	mov word[bx],3	Valid

iii) [4 Mark] For following write the required code, assume that least significant bit is bit number 0:

a) Using one instruction, clear bits 3 and 4: MOV AL, 0xFF and al, 0xE7	b) Using one instruction, toggle bits 2 and 6: MOV AL, 0xAA xor al, 0x44
c) Jump to "Label1" if bit 7 is set, without changing the value of destination operand: MOV AL, 0x80 test al, 0x80 jnz Label1	d) Using one instruction, Take 1's complement of AX: Mov ax, 0x0A0F not ax / xor ax, 0xFFFF

iv) [3 Mark] Given that AX is initialized with **0x4567** and BX is initialized with **0x789A**, what are the values of the flags after executing each of the following instructions and tell whether jump will be taken or not? **Show each and every step to get marks.**

Instruction	ZF	CF	SF	OF	Taken/Not-Taken	Working
add ax, 0x8765 jg l1	0	0	1	0	Not Taken	4567 + 8765 CCCC
sub ax, bx jbe l2	0	1	1	0	Taken	1 4567 - 789A CCCD

Question 2 [CLO 2] [10 Marks]:

i) [5 Mark] You are given memory addresses, their content and values of registers in hexad... would be the value of ax register after executing the instruction given below? Show your w... get full marks.

CS: 0x1D3F, DS: 0x1D3D, SS: 0x1D3E, BP: 0x011F, SI: 0x013F, DI: 0x0112, BX: 0x012D

Instruction: Mov ax, [cs:bx+si+12]

Value of AX after Execution of above instruction: 0x5566

Physical Memory Address	Memory Content (Hex)	Show Your working here:
0x1D666	02EF	012D + 013F = 026C; + C = 0278
0x1D658	035F	Use segment CS
0x1D558	1111	P.A = CS * 10h + offset = 1D3F * 10h + 0278
0x1D668	6655 Little Endian	1D3F0 + 0278 = 1D668
0x1D648	3344	

ii) [5 Mark] The following is an incomplete listing file, answer the following questions. Show your working to get credit.

```
                      [org 0x100]
 00000000             jmp start

 00000003 0500        num1: dw 5
 00000005 0000        square: dw 0

 00000007 A1[0300]    start: mov ax, [num1]
 0000000A 8B0E[0300]  mov cx, [num1]

 0000000E 0106[0500]  loop1:add [square], ax
 00000012 49          dec cx
 00000013 81F90000    cmp cx, 0
 00000017             jg loop1

 00000019 B8004C      mov ax, 0x4C00
 0000001C CD21        int 0x21
```

Working:

a) 0007 − 0003 = 0004

In little endian 0400

E90400

b) 19 − 0E = 0B

2's complement = F5

7FF5

(a) What is the instruction in hexadecimal for the assembly instruction jmp start in the above code, given the opcode of the jmp is 0xE9? E90400

(b) What is the instruction in hexadecimal for the assembly instruction jg loop1 in the above code, given the opcode of the jg is 0x7F? 7FF5

3 [CLO 3] [20 Marks]: Write assembly language program to find Mode of an array i.e. the most frequent 0582
r in the array. Safely assume that input array will be sorted in ascending order and it will always have
tly one mode. Sample run is shown below:

Sample Run:

Example 1	Arr: 1,1,2,2,2,2,4,4,4,4,4	Mode = 4
Example 2	Arr: 1,1,2,2,2,2,3,4,4,4,5	Mode = 2

Solution:

```
[org 0x0100]
jmp start
arr: db 1,1,2,2,2,2,4,4,4,4,4
count: dw 11 ; There are total 11 elements in array
; add extra variables here, if required

start:
```